1. Let $a \neq b$ be two non-zero real numbers. Then the number of elements in the set $X = \left\{z \in C : \mathrm{Re}\left(az^2 + bz\right) = a \text{ and } \mathrm{Re}\left(bz^2 + az\right) = b\right\}$ is equal to

[2023 (06 Apr Shift 2)]

(1) 0

(2) 1

(3) 3

(4) 2

2. For $\alpha,\ \beta,\ z \in \mathbb{C}$ and $\lambda > 1$, if $\sqrt{\lambda - 1}$ is the radius of the circle $|z - \alpha|^2 + |z - \beta|^2 = 2\lambda$, then $|\alpha - \beta|$ is equal to _____.

[2023 (06 Apr Shift 2)]

3. If for $z = \alpha + i\beta,\ |z + 2| = z + 4(1 + i)$, then $\alpha + \beta$ and $\alpha\beta$ are the roots of the equation

[2023 (08 Apr Shift 1)]

(1) $x^2 + 3x - 4 = 0$

(2) $x^2 + 7x + 12 = 0$

(3) $x^2 + x - 12 = 0$

(4) $x^2 + 2x - 3 = 0$

4. Let $A = \left\{\theta \in (0,\ 2\pi) : \frac{1 + 2i\sin\theta}{1 - i\sin\theta} \text{ is purely imaginary}\right\}$ Then the sum of the elements is in A is

[2023 (08 Apr Shift 2)]

(1) 4π

(2) 3π

(3) π

(4) 2π

5. Let the complex number $z = x + iy$ be such that $\frac{2z - 3i}{2z + i}$ is purely imaginary. If $x + y^2 = 0$, then $y^4 + y^2 - y$ is equal to

[2023 (10 Apr Shift 1)]

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) $\frac{3}{4}$

(4) $\frac{4}{3}$

6. Let $S = \left\{z = x + iy : \frac{2z - 3i}{4z + 2i} \text{ is a real number }\right\}$. Then which of the following is **NOT** correct?

[2023 (10 Apr Shift 2)]

(1) $y + x^2 + y^2 \neq -\frac{1}{4}$

(2) $\left(x, y\right) = \left(0, -\frac{1}{2}\right)$

(3) $x = 0$

(4) $y \in \left(-\infty, -\frac{1}{2}\right) \cup \left(-\frac{1}{2}, \infty\right)$

7. Let w_1 be the point obtained by the rotation of $z_1 = 5 + 4i$ about the origin through a right angle in the anticlockwise direction, and w_2 be the point obtained by the rotation of $z_2 = 3 + 5i$ about the origin through a right angle in the clockwise direction. Then the principal argument $w_1 - w_2$ is equal to

[2023 (11 Apr Shift 1)]

(1) $\pi - \tan^{-1}\frac{8}{9}$

(2) $-\pi + \tan^{-1}\frac{33}{5}$

(3) $-\pi + \tan^{-1}\frac{8}{9}$

(4) $\pi - \tan^{-1}\frac{33}{5}$

8. For $a \in \mathbb{C}$, let $A = \{z \in \mathbb{C} : \mathrm{Re}(a + \bar{z}) > \mathrm{Im}(\bar{a} + z)\}$ and $B = \{z \in \mathbb{C} : \mathrm{Re}(a + \bar{z}) < \mathrm{Im}(\bar{a} + z)\}$. Then among the two statements:

$(S1)$: If $\mathrm{Re}(a), \mathrm{Im}(a) > 0$, then the set A contains all the real numbers

$(S2)$: If $\mathrm{Re}(a), \mathrm{Im}(a) < 0$, then the set B contains all the real numbers,

[2023 (11 Apr Shift 2)]

(1) Only $(S2)$ is true

(2) only $(S1)$ is true

(3) Both are true

(4) Both are false

9. Let $S = \left\{z \in \mathbb{C} - \left\{i, 2i\right\} : \frac{z^2 + 8iz - 15}{z^2 - 3iz - 2} \in \mathbb{R}\right\}$. $\alpha - \frac{13}{11}i \in S, \alpha \in \mathbb{R} - \left\{0\right\}$, then $242\alpha^2$ is equal to

[2023 (11 Apr Shift 2)]

10. Let C be the circle in the complex plane with centre $z_0 = \frac{1}{2}(1+3i)$ and radius $r = 1$. Let $z_1 = 1 + i$ and the complex number z_2 be outside circle C such that $|z_1 - z_0||z_2 - z_0| = 1$. If $z_0,\ z_1$ and z_2 are collinear, then the smaller value of $|z_2|^2$ is equal to

[2023 (12 Apr Shift 1)]

(1) $\frac{5}{2}$
(2) $\frac{7}{2}$
(3) $\frac{13}{2}$
(4) $\frac{3}{2}$

11. Let $w = z\bar{z} + k_1 z + k_2 iz + \lambda(1+i), k_1, k_2 \in \mathbb{R}$. . Let $Re(w) = 0$ be the circle C of radius 1 in the first quadrant touching the line $y = 1$ and the $y-$axis. If the curve $Im(w) = 0$ intersects C at A and B, then $30(AB)^2$ is equal to _______.

[2023 (13 Apr Shift 1)]

12. Let $S = \{z \in \mathbb{C} : \bar{z} = i(z^2 + \text{Re}(\bar{z}))\}$. Then $\sum_{z \in S} |z|^2$ is equal to

[2023 (13 Apr Shift 2)]

(1) $\frac{5}{2}$
(2) 4
(3) $\frac{7}{2}$
(4) 3

13. If the set $\left\{Re\left(\frac{z - \bar{z} + z\bar{z}}{2 - 3z + 5\bar{z}}\right) : z \in \mathbb{C},\ \ Re\, z = 3\right\}$ is equal to the interval $(\alpha,\ \beta]$, then $24(\beta - \alpha)$ is equal to

[2023 (15 Apr Shift 1)]

(1) 36
(2) 27
(3) 30
(4) 42

ANSWER KEYS

1. (1)	**2.** (2)	**3.** (2)	**4.** (1)	**5.** (3)	**6.** (2)	**7.** (1)	**8.** (4)
9. (1680)	**10.** (1)	**11.** (24)	**12.** (2)	**13.** (3)			

1. (1)

Given,

$X=\{z \in C: \operatorname{Re}(az^2+bz)=a \text{ and } \operatorname{Re}(bz^2+az)=b\}$

Now, let $z=x+iy$

So, $\operatorname{Re}(az^2+bz)=a$

$\Rightarrow \operatorname{Re}\left(a(x+iy)^2+b(x+iy)\right)=a$

$\Rightarrow \operatorname{Re}\left(a(x^2-y^2+2ixy)+b(x+iy)\right)=a$

$\Rightarrow a(x^2-y^2)+bx=a \quad \ldots$ (i)

And, $\operatorname{Re}(bz^2+az)=b$

$\Rightarrow \operatorname{Re}\left(b(x+iy)^2+a(x+iy)\right)=b$

$\Rightarrow \operatorname{Re}\left(b(x^2-y^2+2ixy)+a(x+iy)\right)=b$

$\Rightarrow \ b(x^2-y^2)+ax=b \ \ldots$ (ii)

From (i) and (ii), (i)$-$(ii) we get,

$(x^2-y^2)(a-b)-x(a-b)=a-b$

$\Rightarrow \ x^2-y^2-x=1 \ldots$ (iii)

From (i) and (ii), (i)+(ii)

$\left((x^2-y^2)+x-1\right)(a+b)=0$ (Note: here $a+b \neq 0$ is considered but it is not clear from the question)

$\Rightarrow x^2-y^2+x-1=0 \ \ldots$ (iv)

Now from (iii) and (iv) we get,

$x=0,\ y^2=-1$ (No solution)

2. (2)

Given,

$\alpha,\ \beta,\ z \in \mathbb{C}$ and $\lambda>1$,

And $\sqrt{\lambda-1}$ is the radius of the circle $|z-\alpha|^2+|z-\beta|^2=2\lambda$

Now we know that equation of circle is given by,

$|z-z_1|^2+|z-z_2|^2=|z_1-z_2|^2$, where radius is given by $r=\frac{|z_1-z_2|}{2}$,

On comparing with $|z-\alpha|^2+|z-\beta|^2=2\lambda$, we get

Radius $\frac{|z_1-z_2|}{2}=\frac{|\alpha-\beta|}{2}$ and $|\alpha-\beta|^2=2\lambda$

$\Rightarrow \frac{|\alpha-\beta|}{2}=\sqrt{\lambda-1}$ (given)

$\Rightarrow |\alpha-\beta|=2\sqrt{\lambda-1}$

$\Rightarrow |\alpha-\beta|^2=4(\lambda-1)$

$\Rightarrow 2\lambda=4(\lambda-1)$

$\Rightarrow \lambda=2$

Hence, $|\alpha-\beta|^2=4 \Rightarrow |\alpha-\beta|=2$

3. **(2)**

Given,

$z=\alpha+i\beta$ & $|z+2|=z+4(1+i)$

Now putting the value of $z=\alpha+i\beta$ in $|z+2|=z+4(1+i)$ we get,

$|z+2|=z+4(1+i)$

$\Rightarrow\sqrt{(\alpha+2)^2+\beta^2}=(\alpha+4)+i(\beta+4)$

Now on comparing real and imaginary part, we get

$\sqrt{(\alpha+2)^2+\beta^2}=\alpha+4\ldots\left(i\right)$ and $\beta+4=0\Rightarrow\beta=-4\ \ldots(ii)$

Now solving,

$\sqrt{(\alpha+2)^2+16}=\alpha+4$

$\Rightarrow\alpha^2+4\alpha+20=\alpha^2+8\alpha+16$

$\Rightarrow\alpha=1$

So, $\alpha+\beta=-3,\ \alpha\beta=-4$

We know that quadratic equation is given by,

$x^2-(\text{sum of roots})x+(\text{product of roots})=0$

So, equation with roots -3 and -4 will be,

$x^2+7x+12=0$

4. **(1)**

Let

$z=\frac{1+2i\sin\theta}{1-i\sin\theta}$

$\Rightarrow z=\frac{1+2i\sin\theta}{1-i\sin\theta}\times\frac{1+i\sin\theta}{1+i\sin\theta}$

$\Rightarrow z=\frac{1-2\sin^2\theta+3i\sin\theta}{1+\sin^2\theta}$

$\Rightarrow z=\left(\frac{1-2\sin^2\theta}{1+\sin^2\theta}\right)+i\left(\frac{3\sin\theta}{1+\sin^2\theta}\right)$

Since, z is a purely imaginary number, so real part must be zero, hence

$\frac{1-2\sin^2\theta}{1+\sin^2\theta}=0$

$\Rightarrow 1-2\sin^2\theta=0$

$\Rightarrow\cos 2\theta=0$

$\Rightarrow\theta=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$, for $\theta\in(0,2\pi)$

$\therefore$Sum of all values $=\frac{16\pi}{4}=4\pi$

5. **(3)**

Given,

The complex number $z=x+iy$ be such that $\frac{2z-3i}{2z+i}$ is purely imaginary,

Now putting the value of $z=x+iy$ in $\frac{2z-3i}{2z+i}$ we get,

$\frac{2(x+iy)-3i}{2(x+iy)+i}=\frac{(2x+i(2y-3))}{(2x+i(2y+1))}\frac{(2x-i(2y+1))}{(2x-i(2y+1))}$

Now taking real part as,

$\frac{4x^2+(2y-3)(2y+1)}{4x^2+(2y+1)^2}=0$

$\Rightarrow 4x^2+4y^2-4y-3=0$

$\Rightarrow x^2+y^2-y-\frac{3}{4}=0$

Now using,

$x+y^2=0\Rightarrow x=-y^2$ we get,

$x^2+y^2-y-\frac{3}{4}=0$

$\Rightarrow y^4+y^2-y-\frac{3}{4}=0$

$\Rightarrow y^4+y^2-y=\frac{3}{4}$

6. (2)

We have,

$Z=\frac{2z-3i}{4z+2i}$

$\Rightarrow Z=\frac{2x+i(2y-3)}{4x+i(4y+2)}$

$\Rightarrow Z=\frac{2x+i(2y-3)}{4x+i(4y+2)}\times\frac{4x-i(4y+2)}{4x-i(4y+2)}$

$\Rightarrow Z=\frac{8x^2+(4y+2)(2y-3)+i[4x(2y-3)-(8xy+4x)]}{16x^2+(4y+2)^2}$

Since, Z is purely real, so $\frac{4x(2y-3)-(8xy+4x)}{16x^2+(4y+2)^2}=0$

$\Rightarrow -12x+8xy-(8xy+4x)=0$

$\Rightarrow x=0$

And,

$4x+i(4y+2)\neq 0$

$\Rightarrow i(4y+2)\neq 0$

$\Rightarrow y\neq -\frac{1}{2}$

Hence this is the required option.

7. (1)

Given,

Two complex numbers w_1 & w_2

$w_1=3+5i$ and $w_2=5+4i$ are both rotated by 90° with respect to origin anticlock-wise and clockwise respectively,

So by concept of rotation we get,

$w_3=iw_1=i(3+5i)=-5+3i$

And $w_4=-iw_2=-i(5+4i)=4-5i$

Now principal argument of $w_3-w_4=-9+8i$ will be $\pi-\tan^{-1}\frac{8}{9}$ {as complex number is in second quadrant so principal argument is given by $\pi-\tan^{-1}\left|\frac{y}{x}\right|$}

8. (4)

Let $a=x_1+iy_1,\ \bar{a}=x_1-iy_1$

And $z=x_2+iy_2,\ \bar{z}=x_2-iy_2$

Now,

$a+\bar{z}=(x_1+x_2)+i(y_1-y_2)$

and $\bar{a}+z=(x_1+x_2)+i(y_2-y_1)$

Now according to the question,

$\text{Re}(a+\bar{z})=x_1+x_2$ and

$\text{Im}(\bar{a}+z)=y_2-y_1$

$A=\{z:x_1+x_2>y_2-y_1\}=\{z:x_1+y_1+x_2>y_2\}$

$B=\{z:x_1+x_2<y_2-y_1\}=\{z:x_1+y_1+x_2<y_2\}$

If $y_2=0$ and $x_1,y_1>0$ then

$A=\{z:x_2>-(x_1+y_1)\}$

A covers a part of negative real axis and therefore, does not contain whole real axis

Similarly if $y_2=0$, and $x_1,y_1<0$, then

$B=\{z:x_2<-(x_1+y_1)\}$

$\therefore B$ covers part of positive real axis and therefore does not cover whole real axis.

Hence, both are false.

9. **(1680)**

Put $z = x + iy$ in the given expression and equate the imaginary part to zero.

$\Rightarrow \frac{z^2+8iz-15}{z^2-3iz-2} = \frac{(x+iy)^2+8i(x+iy)-15}{(x+iy)^2-3i(x+iy)-2}$

$= \frac{x^2-y^2-8y-15+i(2xy+8x)}{x^2-y^2+3y-2+i(2xy-3x)}$

$= \frac{x^2-y^2-8y-15+i(2xy+8x)}{x^2-y^2+3y-2+i(2xy-3x)} \times \frac{(x^2-y^2+3y-2-i(2xy-3x))}{(x^2-y^2+3y-2-i(2xy-3x))}$

$= \frac{(x^2-y^2-8y-15)(x^2-y^2+3y-2)+(2xy+8x)(2xy-3x)+i(2xy+8x)(x^2-y^2+3y-2)-i(2xy-3x)(x^2-y^2-8y-15)}{(x^2-y^2+3y-2)^2-(2xy-3x)^2}$

But $\operatorname{Im}\left(\frac{z^2+8iz-15}{z^2-3iz-2}\right) = 0$,

$\Rightarrow -(x^2 - y^2 - 8y - 15)(2xy - 3x) + (2xy + 8x)(x^2 - y^2 + 3y - 2) = 0$

$\Rightarrow (x^2 - y^2)(2xy + 8x - 2xy + 3x) + (8y + 15)(2xy - 3x) + (2xy + 8x)(3y - 2) = 0$

$\Rightarrow 11x^3 - 11xy^2 + 16xy^2 - 24xy + 30xy - 45x + 6xy^2 - 4xy + 24xy - 16x = 0$

$\Rightarrow 11x^3 + 11xy^2 + 26xy - 61x = 0$

$\Rightarrow 11x^2 + 11y^2 + 26y - 61 = 0$

$\because \alpha \neq 0$

So, put $y = -\frac{13}{11}$, $x = \alpha$

$11\alpha^2 + 11 \cdot \frac{13^2}{11^2} - 26 \cdot \frac{13}{11} - 61 = 0$

$\Rightarrow 121\alpha^2 = 840$

$\Rightarrow 242\alpha^2 = 1680$

Hence this is the required answer.

10. **(1)**

Given,

$z_0 = \frac{1+3i}{2}$, $z_1 = (1 + i)$

So, $|z_1 - z_0| = \sqrt{\left(1 - \frac{1}{2}\right)^2 + \left(1 - \frac{3}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}$

And z_2 be outside circle, $|z_1 - z_0||z_2 - z_0| = 1$

$\Rightarrow \frac{1}{\sqrt{2}}|z_2 - z_0| = 1$

$\Rightarrow |z_2 - z_0| = \sqrt{2}$

Now given, z_0, z_1 & z_2 are collinear,

So, by concept of rotation we get,

$\frac{z_2-z_0}{z_1-z_0} = \frac{|z_2-z_0|}{|z_1-z_0|}e^{i0}$

$\Rightarrow \frac{z_2-z_0}{z_1-z_0} = \frac{|z_2-z_0|}{|z_1-z_0|}(\pm 1)$

$\Rightarrow \frac{z_2-z_0}{z_1-z_0} = \pm 2$

$\Rightarrow z_2 = z_0 \pm 2(z_1 - z_0)$

So, $z_2 = 2z_1 - z_0 = \frac{3}{2} + \frac{1}{2}i \Rightarrow |z_2|^2 = \frac{5}{2}$

Or $z_2 = 3z_0 - 2z_1 = \frac{-1}{2} + \frac{5}{2}i \Rightarrow |z_2|^2 = \frac{13}{2}$

Hence, the smaller value will be, $|z_2|^2 = \frac{5}{2}$

11. (24)

Given,

$w=z\bar{z}+k_1z+k_2iz+\lambda(1+i), k_1, k_2 \in \mathbb{R}.$.

And $Re(w)=0$,

So, taking $z=x+iy$ & $\bar{z}=x-iy$

We get,

$w=x^2+y^2+k_1(x+iy)+k_2i(x+iy)+\lambda(1+i)$

$\Rightarrow Re(w)=0 \Rightarrow x^2+y^2+k_1x-k_2y+\lambda=0$

And, $Im(w)=0 \Rightarrow k_1y+k_2x+\lambda=0$

Now given $Re(w)=0$ circle is touching $y=1$ and $y-$axis, with radius 1, so we have equation of circle

$x^2+y^2-2x-4y+4=0$

So, on comparing with $Re(w)=0$ we get,

$k_1=-2, k_2=4, \lambda=4$

So, $Im(w)=0 \Rightarrow -2y+4x+4=0$

And given $-2\,y+4\,x+4=0$ intersects the circle,

So, solving $x^2+y^2-2x-4y+4=0$ and $-2\,y+4\,x+4=0$ we get,

Intersecting point as $A(0,2)$ and $B\left(\frac{2}{5},\frac{14}{5}\right)$

Hence, by distance formula we get,

$30\left(AB\right)^2=30\left[\left(0-\frac{2}{5}\right)^2+\left(2-\frac{14}{5}\right)^2\right]=24$

12. (2)

Given,

$\bar{z}=i\left(z^2+\mathrm{Re}(\bar{z})\right)$

Now, let $z=x+iy$, so $\bar{z}=x-iy$

Now putting the value in, $\bar{z}=i\left(z^2+\mathrm{Re}(\bar{z})\right)$ we get,

$\bar{z}=i\left(z^2+\mathrm{Re}(\bar{z})\right)$

$\Rightarrow x-iy=i\left(x^2-y^2+2ixy+x\right)$

$\Rightarrow x-iy=i\left(x^2-y^2+x\right)-2xy$

Now comparing real part we get,

$\Rightarrow x=-2xy \Rightarrow x(2y+1)=0$

$\Rightarrow x=0, y=\frac{-1}{2}\,\ldots\ldots(1)$

And imaginary part we get,

$-y=x^2-y^2+x\,\ldots\ldots.(2)$

Now taking Case (I) when $x=0$ in equation (2) we get,

$\Rightarrow -y=-y^2$

$\Rightarrow y^2-y=0 \Rightarrow y=0,1$

So, $z=0,\ i$

Now taking Case (II) when $y=\frac{-1}{2}$ in equation (2) we get,

$\Rightarrow \frac{1}{2}=x^2-\frac{1}{4}+x$

$\Rightarrow x^2+x-\frac{3}{4}=0$

$\Rightarrow 4x+4x-3=0$

$\Rightarrow (2x-1)(2x+3)=0$

$\Rightarrow x=\frac{1}{2},\frac{-3}{2}$

So, $z=\frac{1}{2}-\frac{1}{2}i, \frac{-3}{2}-\frac{1}{2}i$

Now finding, $\sum\left|z\right|^2=0+1+\frac{1}{2}+\frac{5}{2}=4$

13. (3)

Given that the set $\left\{\mathrm{Re}\left(\frac{z-\bar{z}+z\bar{z}}{2-3z+5\bar{z}}\right): z\in\mathbb{C},\ \mathrm{Re}\,z=3\right\}$ is equal to the interval $(\alpha,\beta]$.

Let $z=x+iy$

$=Re\left(\frac{x+iy-(x-iy)+x^2+y^2}{2-3(x+iy)+5(x-iy)}\right)$

$=Re\left(\frac{x^2+y^2+i(2y)}{2+2x-8iy}\right)$

$=Re\left(\frac{(x^2+y^2+2yi)(2(1+x)+8iy)}{(2(1+x))^2+(8y)^2}\right)$

$=\frac{2(x^2+y^2)(1+x)-16y^2}{4(1+x)^2+(8y)^2}$

Now using, $Re(z)=3\Rightarrow x=3$ we get,

$=\frac{8(9+y^2)-16y^2}{64+64y^2}$

$\Rightarrow f(y)=\frac{1}{8}\frac{(9-y^2)}{1+y^2}$

Let $t=\frac{1}{8}\frac{(9-y^2)}{1+y^2}$

$\Rightarrow 8t+8ty^2=9-y^2$

$\Rightarrow y^2(8t+1)+8t-9=0$

Now $D\geq 0$

$\Rightarrow 0^2-4(8t+1)(8t-9)\geq 0$

$\Rightarrow\left(t-\left(\frac{-1}{8}\right)\right)\left(t-\frac{9}{8}\right)\leq 0$

$\Rightarrow t\in\left(\frac{-1}{8},\frac{9}{8}\right]$ as $t\neq\frac{-1}{8}$

Range of $f(y)=(-0.125,1.125]$

$\Rightarrow\alpha=-0.125$

$\Rightarrow\beta=1.125$

$\Rightarrow\beta-\alpha=1.25$

$\Rightarrow 24(\beta-\alpha)=30$

Hence this is the required solution.